

Project 2: Travel Booking Platform using Agile

1. Abstract

The Travel Booking Platform is an integrated web-based solution designed to simplify the process of searching and booking flights, hotels, and holiday packages. Developing this under the Agile methodology allows for the prioritization of features through a "Product Backlog," ensuring that essential functionalities like search filters and secure payment gateways are delivered early in the development lifecycle. The system aims to provide real-time updates on travel availability by integrating with global distribution system (GDS) APIs.

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2. Introduction

- **2.1 Introduction:** This project focuses on building a seamless interface for travelers to manage their entire itinerary in one place.
 - **2.2 Problem Identification:** Users often face high complexity and fragmented data when booking across different platforms, leading to pricing inconsistencies and booking errors.
 - **2.3 Need of the project:** There is a significant demand for a high-performance, mobile-responsive platform that offers dynamic pricing and instant confirmation.
 - **2.4 Project Scheduling:** The project is divided into sprints focusing on: Sprint 1 (API Integration), Sprint 2 (Booking Engine), Sprint 3 (Payment & Security), and Sprint 4 (User Dashboard & Reviews).
 - **2.5 Objectives:** To provide a user-friendly booking experience, ensure 100% secure payment processing, and allow users to manage cancellations effortlessly.
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3. Software Requirement Specification

- **3.1 Purpose:** To digitize and automate the travel agency experience for global users.

- **3.2 Scope:** Covers domestic and international flight bookings, hotel reservations, and rental car services.
 - **3.3 Hardware Requirement / Software Requirement (minimum):**
 - **Hardware:** Core i5 Processor, 8GB RAM, 100GB Disk Space.
 - **Software:** Windows/Linux OS, Java Runtime Environment (JRE), MySQL Database.
 - **3.4 Tools:** Selenium WebDriver for cross-browser testing, Eclipse IDE for development, and Maven for project management.
 - **3.5 Software Process Model:** Agile Scrum Framework, utilizing daily stand-ups and sprint reviews to manage changing requirements.
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4. System Design

- **4.1 Data Dictionary:** Defines tables such as Traveler_Info, Booking_Status, Payment_Log, and Flight_Inventory.
 - **4.2 ER Diagram:** Shows the relationship between User, Ticket, Payment, and Notification.
 - **4.3 DFD (Data Flow Diagram):** Traces the flow of a search query from the user through the API to the database and back as a filtered result.
 - **4.4 System Flow Chart:** Outlines the logic for checking seat availability and processing credit card authorizations.
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5. Implementation

- **5.1 Program Code:** Developed using Java for the backend logic and Selenium for automating the verification of flight search results.
 - **5.2 Output Screens:** Screen captures of the Flight Search Results page, Hotel Selection page, and the final Booking Confirmation receipt.
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6. Testing

- **6.1 Test Data:** Sets of valid and invalid travel dates, international passport numbers, and promo codes.

- **6.2 Test Result: Documentation of successful test passes for "Payment Gateway Redirect" and "Email Confirmation Trigger".**
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7. User Manual

- **7.1 How to use project guidelines: Step-by-step guide on setting up a user profile, selecting a destination, and applying filters.**
 - **7.2 Screen Layouts and Description: Detailed descriptions of icons used for "Instant Booking," "Refundable," and "Amenities".**
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8. Project Applications and Limitations

- **Applications: Applicable for individual travelers, corporate travel desks, and tour operators.**
 - **Limitations: Heavily dependent on the uptime of third-party airline/hotel APIs; dynamic pricing can change during the checkout process.**
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9. Conclusion and Future Enhancement

- **Conclusion: The Agile approach allowed the team to handle complex API integrations effectively while keeping the user interface simple.**
 - **Future Enhancement: Implementation of AI-based travel recommendations based on user history and voice-assisted booking.**
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10. Bibliography & References

- **Includes references to Selenium Documentation, Agile Manifesto, and Java SQL connection guides.**